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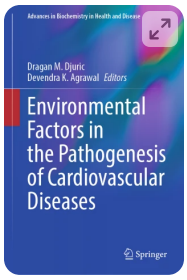

Environmental Factors in the Pathogenesis of Cardiovascular Diseases

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Accessibility Information

Overview

Editors: Dragan M. Djuric, Devendra K. Agrawal

Provides comprehensive and up-to-date information from top-notch investigators in the field

Shows critical discussion on the genetic, environmental, and nutritional factors

Presents prevention and treatment of cardiovascular diseases due to environmental factors

Part of the book series: Advances in Biochemistry in Health and Disease (ABHD, volume 30)

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About this book

Environmental conditions and processes are one of the major pillars on which the human well-being rests. It is the core responsibility of the society to preserve and enhance better conditions for the human well-being. Indeed, there are several evolving unmet needs in public health. Emerging and re-emerging infectious diseases and a surge in the incidence of non-communicable diseases, including cardiovascular diseases (CAD), chronic respiratory diseases, and metabolic diseases have been impediments to sustainable well-being.

Many factors are critical in the global surge in the rate and incidence of cardiovascular diseases. These include the shift from acute to chronic conditions, the shift from single risk factor vs. multiple influences, aging population, global health disparities, exposure to lower harmful influences over a longer period, etc.

However, the epigenetic factors due to unhealthy environment play a most significant role in the underlying pathogenesis of cardiovascular diseases. Unfortunately, this has been ignored for a long time and realized lately to expand and disseminate knowledge to general population, expand research activities to investigate the cellular and molecular mechanisms, and develop better preventive and treatment strategies.

The most significant environmental impoverishment in the pathogenesis of cardiovascular diseases include different genetical, chemical, physical, and biological influences, but not limited to, socio-economic status and lack of nutrients, nutritional aspects including habits, diets and additives, inhaled and ingested pollutants, exhaust gas and gasoline products, tobacco smoke, water pollution, alcohol consumption, soil and mineral pollution, solvents, pesticides, microplastics, non-critical usage of drugs, climate change, extreme atmospheric conditions, extremes in noise and temperature, electromagnetic influences, microwaves and radiation, outdoor light pollution, mental stressors, lack of or over exercise, microbiota and microbiological agents like SARS CoV-2 virus, etc.

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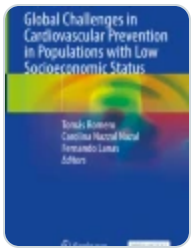

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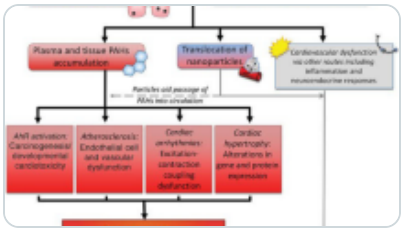

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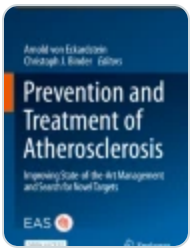

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About the editors

Dr. Devendra K. Agrawal earned MSc, PhD (Biochemistry), PhD (Medical Sciences), MBA and MS (Information Technology & Management). After serving on several academic positions, including Chairman of the Department of Clinical & Translational Science, Senior Associate Dean of Clinical & Translational Sciences, and Senior Vice President for Research and Biotechnology, he is currently the Professor and Director, Department of Translational Research, Western University of Health Sciences, Pomona, CA, USA. Dr. Agrawal is a member of the *Alpha Omega Alpha Honor Medical Society* and is a Fellow of the AAAAI, AHA, APS, and IACS. He has been recognized with many awards, including the highest teaching award, Distinguished Research Career Award, and Distinguished Mentor Award. Dr. Agrawal has served on many grants review panels, including VA Merit Review, DoD, NIH-NIAID, NIH-NHLBI, MRC-UK and US-Israel BSF Grant Review Committees, Science Foundation Ireland, Austrian Science Fund, British Lung Foundation, Swiss NSF, ViCi Netherlands, and Asthma Research Foundation of Western Australia. Dr. Agrawal has served on the editorial board of several prestigious journals. Dr. Agrawal contributed to science by publishing >560 original research articles in peer-reviewed journals and book chapters. He has co-edited six books in Allergy, Asthma, and Immunology, COVID19 and Stem Cells, Flow Cytometry, and this book on the Environmental Factors in the pathogenesis of cardiovascular diseases, and currently working on a book in cardiovascular Toxicity. His major research areas include the cellular, molecular and

immunobiology of diabetes and diabetic foot ulcers, renal diseases, allergy and bronchial asthma, vascular diseases, shoulder rotator cuff injuries, and traumatic brain injury. NIH funds his research projects.

Dr. Agrawal has trained >200 medical students (many of them are now academic physician-scientists), directly supervised and guided 72 graduate students to earn MD-PhD, MD-MS, MS, and PhD, degrees, 74 post-doctoral fellows, 15 assistant professors, many of whom are currently on high positions in academia and industries. Many clinical residents, fellows and research investigators have received training under his direct supervision. He is passionate to train and mentor next generation of clinical and translational researchers."

Dr. Dragan M. Djuric earned MD degree in 1987, MS degree in 1991, PhD degree in 1993, and Clinical Physiologist degree in 2007, all from the Faculty of Medicine University of Belgrade (Belgrade, Serbia). He finished his postdoctoral trainings in Germany (Max Planck Institute for Physiological and Clinical Research and Kerckhoff Klinik GmbH, Bad Nauheim, 1998, 2001-2002), and in USA (Department of Physiology, College of Medicine, University of South Alabama, Mobile, 2000). After serving on several academic positions, including Director and Chair, Institute of Medical Physiology, Faculty of Medicine, University of Belgrade (2006-2012), Founder and Chair, PhD Program in Physiological Sciences (2009-2024), leadership positions in Serbian Physiological Society (2003-2021), FEPS Task Force Group on Education in Physiology (2007-2008), and national atherosclerosis societies (since 1998), he is currently the Professor of Physiology, Faculty of Medicine University of Belgrade (Belgrade, Serbia), and Visiting Professor, Faculty of Medicine, University of Banja Luka (Bosnia and Herzegovina) (since 2019). Dr Djuric is a Fellow of IACS and IUPS Academy of Physiology. He has been recognized with many awards, including the IACS Naranjan Dhalla Award for Innovative Investigators in Cardiovascular Sciences (2024), IACS Oration Award Prof. Harpal Buttar (2024), IACS Howard Morgan Award for Distinguished

Achievements in Cardiovascular Research (2022), IACS James Willerson Award for Excellence in Cardiovascular Sciences (2021), IACS Distinguished Leadership Award in Cardiovascular Sciences (2019), IACS Andras Varro Award for Excellence in Cardiovascular Sciences (2018), IACS Lifetime Achievement Award in Cardiovascular Science, Medicine and Surgery (2015), all by the International Academy of Cardiovascular Sciences (IACS) including European, North American and Indian sections; Serbian Physiological Society Award for Lifetime Achievement in Physiological Sciences (2016); Samuel Racz Medal and Honorary Member for the Contribution in Physiology by the Hungarian Physiological Society (2010); Honorary Member, Bulgarian Society for Cell Biology (2009); Honorary Member, Romanian Society for Laboratory Medicine (2008); Medal of the Yugoslav Society of Cardiology (2002), and Belgrade City October Award (1987). Dr. Djuric has served on the editorial board of several prestigious journals, and in a few grants review panels, and he was a reviewer in journals, books, and meetings abstracts. Dr. Djuric contributed to science by publishing >300 original research, review end editorial articles in peer-reviewed journals and book chapters. He has edited/co-edited 6 books in a field of medical physiology, 7 monographs in a field of endothelium, vascular biology, atherosclerosis and nutrition (including this book on the Environmental Factors in the Pathogenesis of Cardiovascular Diseases, and currently working on a book in Cardiovascular Toxicity), 15 books of abstracts from scientific meetings which he organized/co-organized, and he was guest editor in 13 special/topic issues of prestigious international journals. Dr. Djuric has trained >30 medical students in a research work, and directly supervised and guided 20 graduate students to earn MD-MS and MD-PhD degrees. National Ministry of Education, Science and Technological Development and EU COST funds his research projects. His major research areas include the cardiovascular research, cardioprotection, experimental cardiometabolic models, endothelial dysfunction, experimental vascular models, gasotransmitters and oxidative stress, homocysteine and related vitamins, nutraceuticals and cardiovascular health, risk factors and cardiovascular health,

vascular reactivity and mediators of inflammation, micronutrients deficiencies, atherosclerosis, and environmental factors in the pathogenesis of cardiovascular diseases.

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